

As an Executive Director (ED) on the Cash Equities Central Risk Book (CRB) desk at Nomura, my team relies on robust quantitative modeling to manage risk residuals. When evaluating candidates for a Quantitative Researcher role, I look for a deep understanding of the distinction between **in-sample statistical fit** and **out-of-sample predictive reliability**.

In the high-turnover environment of a CRB, a model that performs well on historical training data but fails to generalize (predict) in production is not just a mathematical failure; it is a financial liability. Below is a deep-dive, line-by-line derivation of the performance metrics you will be expected to defend during a technical interview.

1. In-Sample (IS) Metrics: Measuring Model Fit

In-sample metrics quantify how well the regression hyperplane explains the observed variance in your dataset.

[A] Coefficient of Determination (R^2)

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}} = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

- **Line-by-line derivation:**

1. SS_{res} (**Residual Sum of Squares**): $\sum (y_i - \hat{y}_i)^2$ is the total squared vertical distance between the actual observed returns (y_i) and the model's predicted values ($\hat{y}_i$).
2. SS_{tot} (**Total Sum of Squares**): $\sum (y_i - \bar{y})^2$ is the total variance in the data relative to the sample mean ($\bar{y}$).
3. **The Logic:** The ratio $\frac{SS_{res}}{SS_{tot}}$ represents the proportion of variance *unexplained* by your model. Subtracting this from 1 gives the proportion of variance *captured* (or explained) by the regression.
4. **Note:** If $R^2 < 0$, your model is performing worse than a horizontal line at the sample mean $\bar{y}$.

[B] Adjusted R-Squared ($\bar{R}^2$)

$$\bar{R}^2 = 1 - \left[\frac{(1 - R^2)(n - 1)}{n - K - 1} \right]$$

- **Derivation:** Standard R^2 artificially increases every time you add a new factor (K), even if the factor is noise.
- **The Penalty:** The term $(n - 1)/(n - K - 1)$ scales the error penalty. As the number of regressors (K) approaches the number of observations (n), the denominator $(n - K - 1)$ decreases, causing the fraction to grow and reducing $\bar{R}^2$. This enforces **parsimony** in factor selection.

[C] Information Criteria (AIC & BIC)

$$AIC = 2K - 2\ln(\hat{L}), \quad BIC = K\ln(n) - 2\ln(\hat{L})$$

- **Step-by-step:** 1. $\ln(\hat{L})$ is the log-likelihood of the model. A higher likelihood implies a better fit.
- 2. $2K$ or $K\ln(n)$ are penalty terms for parameter count.

3. **BIC vs. AIC:** The BIC uses $\ln(n)$ instead of 2 as the multiplier for K . In large CRB datasets where n is very large, BIC penalizes complexity much more heavily than AIC, ensuring that we only retain factors that have genuinely high signal-to-noise ratios.

2. Out-of-Sample (OOS) Metrics: Measuring Generalization

Predictive models for the CRB must be evaluated on unseen data to prevent **look-ahead bias** and **overfitting**.

[A] Out-of-Sample R-Squared (R_{OOS}^2)

$$R_{OOS}^2 = 1 - \frac{\sum_{t \in OOS} (y_t - \hat{y}_t)^2}{\sum_{t \in OOS} (y_t - \bar{y}_{train})^2}$$

- **Line-by-line Explanation:**

1. **Numerator:** We sum the squared errors of the model on the *test* (OOS) period.
2. **Denominator:** We measure the variance of OOS data relative to the **mean of the training set** ($\bar{y}_{train}$).
3. **Crucial Principle:** We do not use the mean of the test set ($\bar{y}_{test}$) here. Using $\bar{y}_{test}$ would introduce "leakage" from the future.
4. **Why negative?** If our model's forecast $\hat{y}_t$ is less accurate than simply predicting that next month's return will equal the historical average return of the training period, R_{OOS}^2 will turn negative.

[B] Information Coefficient (IC)

$$IC = \text{Corr}(\hat{y}_t, y_t)$$

- **Theoretical Application:** In the CRB, we often care more about the *rank order* of our predicted returns than the absolute magnitude. The IC measures the strength of the linear relationship between predicted and realized returns. For a researcher on my desk, an *IC* of 0.05 is often significantly more valuable than a high R^2 , as it suggests our signal has persistent, reliable information.

3. Application: Theory & Code

To evaluate a factor risk model, we implement a **walk-forward validation** loop. This avoids the mistake of testing on the entire dataset at once.

```
import numpy as np

def evaluate_factor_model(y_true, y_pred, y_train_mean):
    """
    Computes OOS metrics for a factor return prediction model.
    y_true: Realized returns in the OOS window.
    y_pred: Predicted returns from the model.
    y_train_mean: Mean of the returns during the training period.
    """
    # SS_res: Squared deviations of model prediction from realization
    ss_res = np.sum((y_true - y_pred)**2)
```

```
# SS_tot: Squared deviations of realization from training mean
# Note: We use y_train_mean to ensure strict OOS evaluation
ss_tot = np.sum((y_true - y_train_mean)**2)

r2_oos = 1 - (ss_res / ss_tot)
ic = np.corrcoef(y_true, y_pred)[0, 1]

return r2_oos, ic
```

CRB Follow-up: If you find that your model has a high IS- R^2 but a negative OOS- R^2 , you have almost certainly overfit to historical noise (e.g., market-wide trends that do not persist). The fix is to apply **shrinkage methods** (like Lasso or Ridge regression) or **dimensionality reduction** (PCA) to limit the number of active parameters K .

4. Summary for the ED Interview

If asked which metric is most important for a factor risk model, do not choose just one. State:

1. Use **BIC** in-sample to ensure the model is parsimonious.
2. Use **OOS- R^2** to confirm that the model actually possesses predictive utility (a high bar in equity markets).
3. Use **IC** to ensure the model correctly ranks the idiosyncratic risk residuals, which is vital for the desk's relative-value trading.